

1. I used Chatgpt and Gemini for the codes that I generated.
2. I used google to search up each of the AI generators on the lab computer that I am using.

Objective

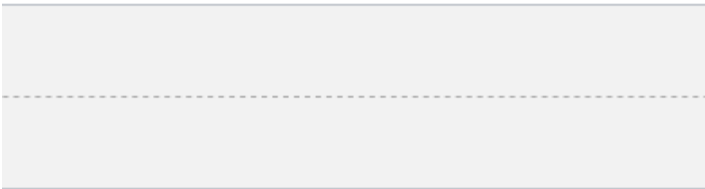
Write a C program that picks a secret number between 1 and 10.

The user gets up to 3 tries to guess the number.

Requirements

- Use a fixed secret number (for example, 7).
- Use `printf` to prompt the user and `scanf` to read their guess.
- Use a `while` or `for` loop to let the user guess up to 3 times.
- Use `if-else` statements to check if the guess is too high, too low, or correct.
- Stop the loop immediately if the user guesses the correct number.
- Print a final win or lose message.

Example Output



text

```
Guess a number between 1 and 10.  
Attempt 1/3. Enter your guess: 5  
Too low! Try again.  
Attempt 2/3. Enter your guess: 8  
Too high! Try again.  
Attempt 3/3. Enter your guess: 7  
Correct! You win!
```

3. This is the prompt I used for both generative ai's I used.

```

#include <stdio.h>

int main() {
    int secret = 7;
    int guess;
    int attempt = 1;
    int correct = 0;

    printf("Guess a number between 1 and 10.\n");

    while (attempt <= 3) {
        printf("Attempt %d/3. Enter your guess: ", attempt);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            correct = 1;
            break;
        } else if (guess < secret) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }

        attempt++;
    }

    if (!correct) {
        printf("You lose! The secret number was %d.\n", secret)
    }

    return 0;
}

```

4.

Chatgpt

this is the code I got from

```

#include <stdio.h>

int main() {
    int secret = 7;
    int guess;
    int max_attempts = 3;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= max_attempts; i++) {
        printf("Attempt %d/%d. Enter your guess: ", i, max_attempts);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            won = 1;
            break;
        } else if (guess < secret) {
            if (i < max_attempts) {
                printf("Too low! Try again.\n");
            }
        } else {
            if (i < max_attempts) {
                printf("Too high! Try again.\n");
            }
        }
    }

    if (!won) {
        printf("You lose! The secret number was %d.\n", secret);
    }

    return 0;
}

```

This is my code from Gemini.

5. For correctness on the code with chatgpt, when I had typed in "Hotdog" the code ran through all the attempts and ran the code for if you used too many attempts, the code did not seem to crash and stop, however, it shouldn't automatically just drop all the attempts. Gemini also did the same thing as Chatgpt so both are not doing well in the sense of not breaking after a non-number is entered into the code. Neither have any lines for errors.

For time complexity both do well and only takes less than a second for the code to run through, I would say that if there would be a bug then a for loop would be better to use since there would be no chance for a for loop to continuously run since a while loop could continue on forever but a for loop stops at what the parameters have.

Space complexity, I would definitely go with a for loop(Gemini) since it is for sure set and a while loop could continue on in the case of a bug allowing it to continue on.

Maintainability, both are incredibly similar as the only difference here is the for in loop, both do not have any comments either

6. I would choose Gemini mainly because of the for loop.
7. For fixing the Gemini coding, I added an error for if used did not enter in a number, then they would not get penalized for using a string, and get to continue guessing. Then I added comments for maintainability so it would be easier to change if needed in the future.

Prologue:

Name: EECS 348 Assignment 1

Desc.: This program is a game that a user get to guess a secret number between 1-10, and then loops 3 times and after the third guess, if not guessed correctly, will tell the user if they lost.

Inputs: Any number between 1 and 10, any other answers will be treated as an error.

Outputs: "Attempt %d/%d. Enter your guess: "

"Error: Please enter a number.\n"

"Correct! You win!\n"

"Too low! Try again.\n"

"Too high! Try again.\n"

"You lose! The secret number was %d.\n"

Code Sources: Chatgpt and Google Gemini

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